

GRADUATE SCHOOL, PROFESSIONAL, AND LIFE
CHOICES: AN OUTCOME ASSESSMENT CONFIRMATION
STUDY MEASURING POSITIVE STUDENT OUTCOMES
BEYOND STUDENT EXPERIENCES FOR PARTICIPANTS IN
COMPETITIVE INTERCOLLEGIATE FORENSICS

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Participation in competitive, intercollegiate forensics is a unique educational opportunity that teaches direct and indirect skills with educational and societal implications. Hundreds of studies over the past nine decades have attempted to examine exactly what competitive debate is, what it should teach and to whom, and what benefits, if any, our student competitors should look forward to as they graduate and exit the competitive experience. In a meta-analysis conducted by Rogers (2002) of 682 forensic articles, books, conference proceedings, and organizational reports, 657 credited debate participation with having positive outcomes consistent with a wide variety of educational, social and behavioral goals.

According to a representative sample of these studies, competitive debate is credited with increasing critical thinking skills (Beckman, 1957; Brembeck, 1949; Colbert, 1987; Cross, 1961; Gruner, Hussman & Luck, 1971; Hill, 1993; Horn & Underberg, 1993; Howell, 1943; Husman, Ware & Gruner, 1972; Jackson, 1961; Rowland, 1995; Williams, 1951; Williams et al, 2001), increasing public presentational skills (Colbert & Biggers, 1985; Millsap, 1998; Stenger, 1999; Williams et al, 2001); teaching public advocacy and social responsibility (Bartanen, 1998; Brand 2000; Brownlee, 1978; Derryberry, 1998; Williams et al., 2001) and offering excellent professional training (Colbert & Biggers, 1985; Hill, 1983; Schneider, 1984; Spangle & Knapp, 1996). It also increases knowledge, self-

confidence, poise, and a wide range of skills necessary for academic success (Bartanen, 1998; Colbert & Biggers, 1985; Derryberry, 1998; Hill, 1983; Jones, 1994; Williams et al., 2001).

In addition, forensic educators argue that debate teaches social responsibility and advocacy (Bartanen & Frank, 1994; Freeley, 1996; Hollihan & Baaske, 1994; Jones, 1994; Rowland, 1995) and enhances a student's academic and professional abilities (Carleton, 1949; Colbert & Biggers, 1985; Derryberry, 1998; Hill, 1983; Jones, 1994; Level, 1957; Pratt, 1990; Schneider, 1984; Spangle & Knapp, 1996; Stenger, 1999; Walker, 1971; Williams, et al 2001). Researchers (Bartanen, 1998; Derryberry, 1998; Millsap, 1998) have concluded that debate teaches important leadership skills. As Bartanen (1998) concludes: "debate fosters leadership skills of reflection, connectedness and advocacy. Forensics programs are valuable models of learner-centered pedagogy and are underutilized resources for diversity education on the liberal arts campus" (pp. 12-13). Susan Millsap (1998) goes so far as to say that a forensics education alone has the ability to be pedagogically effective across the liberal arts curriculum.

These are bold claims, to be sure, but claims that the majority of the readership of this article would probably support for the majority of their forensics students. However, the literature offers little in terms of quantitative evidence to support the positive outcomes for participation that it claims. The majority of the research to date offers random "snapshots" of performance based largely upon anecdotal evidence that fails to offer valid comparisons to non-debate control groups. More than thirty years ago, Andersen argued that "[I]n an age of educational accountability, the forensic community is and will increasingly be called upon to tell what it seeks to do, how well it accomplishes its goals, and what other effects it has. Surprisingly, there seems little interest in such research at this time" (1974, p. 155). Little has changed in the literature to address Anderson's concerns over the past three decades.

What is needed for administrators, fellow academicians, and parents is quantitatively based assessment data that tracks, evaluates, and assesses populations of both *debate* and

non-debate students highlighting significant differences between the two populations as they matriculate through their college careers and into the post-graduation world. Differences in outcomes at various junctures in the longitudinal study period could be directly attributed to the differences in the study populations.

The purpose of the original study published in 2002, hereafter referred to as *Study One*, and this follow up study, referred to as *Study Two*, was and is to provide quantitative support for the conclusion that participation in competitive, intercollegiate forensics is directly correlated with positive educational, social, and behavioral outcomes as compared to outcomes for non-debate students. A summary of *Study One* is helpful to understand fully the background and scope of *Study Two*.

Study One

During the Summer of 1997, twenty-eight Directors of Forensics were asked to identify incoming first-year students who had been recruited to participate on their forensics teams and twenty non-debate, first-year student volunteers recruited from their basic teaching load. A basic intake survey was distributed to a total of 760 first-year students (563 non-debate and 197 debate). The survey was designed to minimize demographic, academic, extracurricular and social differences between the two groups. 100 *debate* and 100 *non-debate* students were eventually selected for inclusion in the four-year study. For a complete discussion of the methodology of the selection criteria and process see Rogers (2002).

An 84-item survey instrument was developed and a five-factor solution confirmed by using orthogonal factor analysis with varimax rotation and then subjecting those factors to ordinary least squares confirmatory factor analysis as discussed in the work of Hunter and Cohen (1969). The instrument used a 7-point Likert scale and was collected at the end of each study year (May 1998, 1999, 2000, 2001). Natural attrition over the course of the study resulted in a final N of 154 students for the complete four-year comparison (debate = 73; non-debate = 81).

At the conclusion of each year, all data were collated and analyzed. The data set was divided into *debate* and *non-debate* groupings and then subjected to statistical analysis to look for statistically significant differences between groups in five critical outcomes: 1) *social responsibility*; 2) *cultural understanding and tolerance*; 3) *academic success*; 4) *moral and ethical issues*; and 5) *psychological multipliers*. All differences between the groups were expressed in Pearson Correlations.

The first area, *social responsibility*, was represented by a cluster of seventeen questions separated into four distinct measures: 1) voting; 2) social volunteerism; 3) political volunteerism; and 4) participation in social advocacy. Within this outcome, *debate* respondents experienced much higher positive outcomes than their *non-debate* peers. All four comparisons were statistically significant. Debate students were much more likely to consistently vote in political elections. The positive correlation between debate and political volunteerism was strong. Over three-quarters of the debate students were directly involved in political campaigns, party work, and/or student organizations. Additionally, debate students were more likely to hold positions of leadership in those organizations (Mean over 4 years = 43%). Finally, debaters were moderately more likely to be involved in social advocacy; that is, by using their speaking/persuasive skills in direct attempts to influence social activism and policy outcomes.

The second area, *cultural understanding and tolerance*, was represented by a cluster of eighteen questions separated into four distinct comparisons: 1) cross-cultural coursework; 2) cross-cultural relationships; 3) cross-cultural-organization (defined as membership in cross-cultural organizations); and 4) rejection of social norming (defined as the rejection of society's absolute right to determine "correct" behavior. In effect, does the student have the sophistication to understand differing attitudes towards social norming?). Two comparisons emerged as statistically significant. *Debate* students were slightly more likely to enroll in cross-cultural coursework. The most significant positive difference noted between respondent groups was within the outcome of *Social Norming*. Debate students were much more likely to reject classical definitions of social norming.

The third area, *academic success*, was represented by a cluster of nine questions targeting six comparisons: 1) GPA; 2) Propensity to matriculate on time; 3) propensity to change majors; 4) professional internship involvement; 5) acceptance into graduate programs; and 6) job offers upon graduation. No significant differences were found between groups in the propensity to graduate on time or change majors. In contrast, there is a strong correlation between debate experience and involvement in professional internships that supported the student's major course of study. This involvement may be linked to a moderately high correlation between debate and job offers at date of graduation rates. Finally, a strong correlation between debate experience and acceptance into graduate programs and professional schools was reported. In summary, debate experience has definite positive pre and post-graduation, "real world" outcomes. Just as important, the negative perceptions of lower GPA's, slower matriculation and graduation, and unfocused wandering from major to major are not statistically supported for the *debate* study population when compared to their *non-debate* peers.

The fourth area, *moral and ethical issues*, was represented by a cluster of fifteen questions targeting four specific comparisons: 1) Propensity to distort the truth; 2) Belief in situational ethics; 3) Propensity to ignore conflicting evidence; and 4) Belief in the just society tradition. No significant differences were reported between the subject groups in the areas of propensity to distort the truth or a belief in situational ethics. Again, the data were perhaps more interesting for what they did not report. Any inference that debaters are more likely than their peers to distort the truth or to resort to situational ethics to "win the point" received no statistical support. An overall positive outcome assessment for debate experience with regard to propensity to ignore conflicting evidence and a belief in a just society tradition was reported. Debaters are significantly less likely to ignore evidence which conflicts with what they believe and are significantly more comfortable with society's system of justice. There appears to be a stronger belief in the ability of concerned citizens to make a difference on the part of the *debate* study group as compared to their *non-debate* peers.

The fifth area, *psychological multipliers*, was represented by twenty-five questions targeted at six comparisons: 1) Feelings of depression/anxiety; 2) Overwhelmed under pressure; 3) Self-confidence; 4) Confidence in communication skills; 5) Ability to maintain long-term relationships; and 5) Propensity for flexibility. Five positive outcomes for those who participate in debate were reported: 1) debaters reported significantly less feelings of depression and anxiety; 2) significantly less feelings of being overwhelmed under pressure; 3) significantly higher feelings of confidence and positive outlook; 4) significantly higher confidence in their ability to communicate feelings and positions through oral and written communication; and 5) are significantly more flexible in their outlook on life. The only area where the study groups exhibited no significant differences was in their ability to maintain successful long-term relationships. The *debate* respondents found it slightly more difficult to maintain long-term relationships.

A consolidated comparison table between *debate* and *non-debate* groupings was offered in the original work and is presented here as Table 1.

Table 1
Consolidated Comparison between Debate and Non-Debate Groups
Study One – Undergraduate Comparisons

	Study Populations	
	Debate Students	Non-debate Students
Five Critical Outcomes		
Social Responsibility		
Propensity to vote	+0.617	+0.113
Propensity towards Social Volunteerism	+0.237	+0.198
Propensity towards Political Volunteerism	+0.818	+0.101
Propensity for Participation in Social Activism	+0.837	+0.459
Cultural Tolerance and Understanding		
Enrollment in Cross-Cultural Coursework	+0.259	+0.193
Maintain Cross-cultural relationships	No significant difference	
Involvement in Cross-Cultural Organizations	No significant difference	
Reject Classical definition of social norming	+0.817	+0.391
Academic Success		
Propensity to maintain 3.5 GPA	+0.753	+0.697
Propensity to Matriculate on time	No significant difference	
Propensity to change major	No significant difference	
Propensity for Professional Internship	+0.081	+0.253
Acceptance to Graduate Program	+0.715	+0.261
Job Offers in Major at Graduation	+0.537	+0.398
Moral / Ethical Outcomes		
Propensity to Distort the Truth	No significant Difference	
Belief in Situational Ethics	No significant difference	
Propensity to Ignore Conflicting Evidence	+0.113	+0.817
Belief in the Just Society Tradition	+0.870	+0.596
Psychological Multipliers		
Propensity for Depression / Anxiety	+0.097	+0.413
Feeling Overwhelmed under Pressure	+0.011	+0.211
Feelings of Self-Confidence / Positive Outlook	+0.799	+0.700
Confidence in Communication Skills	+0.873	+0.417
Maintain long-term relationships	No significant difference	
Propensity for Flexibility	+0.537	+0.336

Note: All positive behavioral comparisons appear in bold face.

The conclusion reached by *Study One* was clear: in almost every case, in almost every area examined, participation in debate engendered significant positive outcomes for the *debate* group as compared to those respondents from the *non-debate* group. Therefore, for an undergraduate population, participation in intercollegiate competitive debate had educational, social and behavioral benefits that were not experienced by the non-debate comparison group. The question is whether these benefits extend beyond the undergraduate experience. Rogers (2002) suggested that “it would be interesting to follow the populations ‘post graduation’ to see if the positive outcomes reported were long lasting or dissipated over time. Do the outcomes of debate participation have life-long results that continue to engender overall success?” (p. 23). To follow up on the previous research, *Study Two* was launched in Summer 2001. The intent of *Study Two* was to extend the original research to answer

RQ1: Will significant positive differences in the five critical areas of: academic success; social responsibility; psychological multipliers; cultural tolerance and understanding; and moral/ethical issues as reported in Study One continue for the debate study population in their post-graduation experiences?

In addition to investigating the continued validity of positive differences in the five critical outcomes, an additional research question is advanced. Because many of the subjects from both respondents pools, *debate* and *non-debate*, were moving into their first job after graduation,

RQ2: Are there differences in performance between debate and non-debate groups in their career paths post-graduation?

Study Two

Subjects

Upon completion of the final 2001 survey and graduation at the close of the Spring and Summer terms of 2001, all respondents were provided with a self-addressed, stamped contact information card and asked to keep in touch with the research team. In early Fall

2001, the respondents were asked to continue their participation in the study for an additional four years. Attrition from the respondent pool resulted in a final N of 119 for the four-year, post-graduation comparisons (68-*debate*; 51-*non-debate*). Roughly 60% (.595%) of the original 200 respondents completed the full eight-year study. As in *Study One* all data from mid-study dropouts was excluded from the final comparison analysis.

Methodology

Since this study is longitudinal in nature, as it seeks to track respondents of the original *Study One* into the post-graduation world for an additional four years, only minor changes were required in the methodology for *Study Two*. The 84-item, previously validated, survey instrument used in *Study One* was administered at the conclusion of each year (May 2002, 2003, 2004, 2005). The goal was to collect and analyze data for significant differences between the *debate* and *non-debate* respondent pools in the critical outcomes previously identified and tracked through their undergraduate experiences: 1) social responsibility; 2) cultural understanding and tolerance; 3) academic success; 4) moral and ethical issues; and 5) psychological multipliers. In addition, anecdotal data was collected that asked the respondents about their initial work experiences.

Four of the critical outcomes remained relevant for all respondents: (1) social responsibility; 2) cultural understanding and tolerance; 4) moral and ethical issues; and 5) psychological multipliers. Therefore, as in *Study One*, all respondents' data were included in those comparisons. However, the critical outcome of *academic success* was only relevant for those respondents who had continued their academic careers into graduate and professional schools. Therefore, only those respondents who continued their academic careers were compared under critical outcome 3) *academic success* (N=66 successfully completed by May 2005: *debate*=46; *non-debate*=20).

After the fourth year (2005), all data were collated and analyzed. Though the 84-item survey instrument had been previously validated on this respondent pool, to ensure that no significant changes had occurred, the scores in each of the five critical outcome areas were

averaged and compared using paired *t-tests*, which is similar to a one-sample *t-test* on differences ($H:d=0$ v $H_a=0$). The higher the number of comparisons made, the greater the risk of a Type 1 error. To protect the integrity of the process, Bonferroni's approach to multiple comparisons was used (.05 divided by 2 times the number of comparisons ($N=119$ – critical outcomes 1, 2, 4, & 5 and $n=66$ for critical factor 3 = $\text{Prob} > |T| = .0002$ and $.0003$ for critical factor 3). As a result, any comparison in critical outcomes 1, 2, 4, & 5 where $p < .0002$ was considered statistically significant. For critical outcome 3, a comparison of $p < .0003$ was considered statistically significant. SPSS was used to analyze and manipulate the data set. Those results are reported in Table 2.

Table 2

Response Comparisons for Statistical Significance

	Critical Outcomes				
	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
	Social	Cultural	Academic	Moral / Ethical	Psychological
	Responsibility	Tolerance	Success		Multipliers
Respondent Group					
Debate	+1.517	+1.445	+1.771	+1.190	+0.911
Prob > T	0.0001	0.0001	0.0001	0.0001	0.0001
Non-debate	+1.231	+0.513	+1.271	+0.489	+0.391
Prob > T	0.0001	0.0001	0.0001	0.0001	0.0001

All values represent means. Positive values equal positive relationships.

Table 2 does not report correlated relationships, but simply significant differences in the way the subject groups responded to the statements measuring the five critical outcome areas, the obvious differences in perception are interesting. At the very least, the two respondent groups perceive of themselves and their performance in the five critical outcomes measured quite differently.

Results and Discussion

As in *Study One*, the responses of each respondent group for each of the five critical outcomes were examined using Pearson correlations. *Study One* concluded that in almost every case, in almost every area and sub-grouping compared, participation in debate had significant positive outcomes for the respondent population as compared to their *non-debate* peers. This study compares entry into and success within graduate school programs, law and professional schools, first career moves and promotions, continued social responsibility and participation, and whether the positive outcomes identified in *Study One* continue to be supported in the former *debate* respondent pool.

Critical Outcome 1 – *Social Responsibility*

Study One reported that the *debate* respondents experienced much higher positive outcomes than their *non-debate* peers in all four areas of comparison during their undergraduate years. The results from this study are even more dramatic. During the four years following graduation, the positive outcome gap in *social responsibility* between the *debate* participants and their *non-debate* peers became even more pronounced. All comparisons were statistically significant. For ease of comparison, *Study One* and *Study Two* Pearson correlations in each of the four sub-assessments are offered side-by-side and reported in Table 3.

Table 3

8-year Consolidated Comparison between Debate and Non-Debate Groups
Social Responsibility

	<i>Study One</i> Undergrad 1997-2001		<i>Study Two</i> Grad & Beyond 2001-2005	
	Debate	Non-debate	Debate	Non-Debate
Critical Outcome 1: Social Responsibility				
Propensity to vote	+0.617	+0.113	+0.760	+0.121
Propensity towards Social Volunteerism	+0.237	+0.198	+0.311	+0.175
Propensity towards Political Volunteerism	+0.818	+0.101	+0.762	+0.079
Propensity for Participation in Social Activism	+0.837	+0.459	+0.812	+0.215

Former *debate* participants were much more likely than their *non-debate* peers to consistently vote in political elections. Voter participation for both *debate* and *non-debate* respondents increased after graduation. However, the increase for former debate participants was much higher; in effect, widening the gap between groups. The *debate* respondents increased their participation in *social volunteerism* and reported slight decreases in both *political volunteerism* and participation in *social activism*. During the same four-year period, *non-debate* respondents reported a modest decrease in *social volunteerism*, but dramatic reductions in both *political volunteerism* and *social activism*.

In summary of the outcome *social responsibility*, former *debate* participants reported increased correlations to positive outcomes in behavior when compared to their *non-debate* peers. In the four years after graduation, the *debate* group was much more likely than their *non-debate* peers to continue to act in a way that reflected the acceptance of social responsibility through voting, to volunteer their time to participate in social and political campaigns, and to use their skills to advocate social change.

Critical Outcome 2 – *Cultural Understanding and Tolerance*

In *Study One*, two comparisons emerged as statistically significant. *Debate* respondents were slightly more likely to enroll in cross-cultural coursework and were much more likely to reject classical definitions of social norming. No significant differences were noted in either group's propensity to maintain intercultural relationships or to be involved in cross-cultural organizations. Direct comparisons between *Study One* and *Study Two* are reported in Table 4.

Table 4
8 year Consolidated Comparison between Debate and Non-Debate Groups
Cultural Tolerance and Understanding

	<i>Study One</i> Undergrad 1997-2001		<i>Study Two</i> Grad & Beyond 2001-2005	
	Debate	Non-debate	Debate	Non-Debate
Critical Outcome 2: Cultural Tolerance and Understanding				
Enrollment in Cross-Cultural Coursework	+0.259	+0.193	+0.357	+0.226
Maintain Cross-cultural relationships	No significant difference		+0.259	+0.131
Involvement in Cross-cultural Organizations	No significant difference		+0.352	+0.210
Reject classical definition of social norming	+0.817	+0.391	+0.739	+0.219

During the four-year follow-up period, former *debate* respondents reported significant differences in behavior, once again widening the positive outcome gap between themselves and their *non-debate* peers. Of those students enrolled in graduate programs (N=66), the *debate* group was much more likely to enroll in cross-cultural coursework. The *debate* group's positive correlation with the sub-assessments of *cross-cultural relationships* and *membership in cross-cultural organizations* strengthened during the four-year follow-up

period. During the same period, the *non-debate* group's correlations registered slight declines in both areas. Thus, what was not reported as a significant difference between the groups during their undergraduate years emerged as significant differences in behavior post-graduation. Both groups reported weaker positive correlations with the tendency to *reject social norming*. However, the difference between groups is still significant.

In summary of the examination of critical outcome 2: *cultural tolerance and understanding*, in each of the four sub-assessment scales, former *debate* participants reported increased correlations to positive outcomes in behavior when compared to their *non-debate* peers. In the four years after graduation, the *debate* group was much more likely than their *non-debate* peers to continue to act in a way that reflected social tolerance and understanding of cultural identities other than their own through the maintenance of intercultural relationships, membership in intercultural organizations, enrollment in cross-cultural course-work, and the rejection of social norming.

Critical Outcome 3 – *Academic Success*

Study One reported that the *debate* grouping held a slight edge in maintaining a GPA of 3.5 or better than their *non-debate* peers. No statistically significant differences were found in terms of *matriculating on time* or *changing majors* between study groups. However, *debate* respondents were much more likely to participate in professional internships related to their majors, to be accepted into graduate school, and to have a job offer upon graduation than their *non-debate* peers. In the following comparisons for *Study Two*, only those respondents enrolled in a professional or graduate school are included. Comparisons between studies and groups in academic success are reported in Table 5.

Table 5

8 year Consolidated Comparison between Debate and Non-Debate Groups
Academic Success

Critical Outcome 3: Academic Success	<i>Study One</i> Undergrad 1997-2001		<i>Study Two</i> Grad & Beyond 2001-2005	
	Debate	Non-debate	Debate	Non-Debate
Propensity to maintain 3.5 GPA	+0.753	+0.697	+0.911	+0.793
Propensity to Matriculate on time	No significant difference		+0.919	+0.801
Propensity to change major	No significant difference		No significant difference	
Propensity for Professional Internship	+0.081	+0.253	No significant difference	
Acceptance to Graduate Program	+0.715	+0.261	NA	
Job Offers in Major at Graduation	+0.537	+0.398	+0.915	+0.511

Of the 58 *debate* subjects accepted into a graduate or professional school at the close of *Study One* (2001), 53 expressed an intention to continue. This study began with 51 of those respondents. Twenty-nine were accepted into Master's degree programs. Eighteen were accepted in law school. Four were accepted into professional schools. This study ended with 46 subjects successfully completing their post-graduation academic career with the awarding of 26 Master's degrees, sixteen successfully passed B.A.R. examinations, and four completed professional schools (certified accountants, computer programmers, military officer's training). As of May 2005, the remaining 5 *debate* respondents had either dropped out (2) or were not finished but still enrolled (3).

Of the 28 *non-debate* subjects accepted into a graduate or professional school at the close of *study one*, this study began with 26. Twenty had been accepted in graduate schools with the remaining 6 being split between law school (4) and medical school (2). This study concluded with 20 *non-debate* subjects successfully completing their post-graduation academic careers: 15 Master's degrees, four passing the B.A.R, and one M.D. Of the

remaining six *non-debate* subjects, three had dropped out, two were still enrolled but not finished, and one was deceased.

The successful academic completion rates are 90.2% for the *debate* grouping and 76.92% for the *non-debate* group. A cursory examination of this study group would seem to indicate that the *debate* group was somewhat more successful in terms of completing their graduate programs than their *non-debate* peers. In addition, the *debate* group reported higher scores on the LSAT and GRE than their *non-debate* peers. The LSAT (Law School Admission Test) contains two logical reasoning sections with between 24 and 26 questions that analyze statements for errors in logic. In addition, there is an analytical section made up of 24 questions that seek to measure the respondents' ability to solve complex deductive puzzles. The *debate* group scored an average of 5.5 points higher on the two logical reasoning sections and 3 points higher on the analytical section for an increase, on average, of 8.5 points on the LSAT over their *non-debate* peers (debate: 168; non-debate: 159).

The *debate* group also outperformed their *non-debate* peers on the GRE (Graduate Record Examination). The GRE is comprised of three sections: verbal, quantitative, and analytic writing. The verbal section is designed to test the respondents' ability to understand and analyze written material. The verbal section is worth between 200 and 800 points. The *debate* group averaged a score of 687 on the verbal section while the non-debate group's averaged a score of 630. Quantitative scores were similar: *debate* 583; *non-debate* 574. However, on the final section, analytic writing, designed to test the respondents' ability to examine claims and accompanying evidence and to support ideas with relevant reasons and examples, the *debate* group averaged a 5.1 (scores range from 0-6) dramatically outperforming the *non-debate* peers who averaged a score of 3.7.

In summary of the examination of the critical factor of *academic success*, the debate group was much more likely to maintain a GPA above a 3.5 while enrolled in their graduate or professional studies. Although there was no statistical difference in either group to matriculate on time as undergraduates, graduate *debate* subjects were much more likely to finish their graduate studies on time. Finally, the *debate* respondents had a much higher *job*

offer at graduate rate than their *non-debate* peers. The authors are pleased to report that eight new forensic coaches were graduated from the ranks of the *debate* respondents.

Hopefully, these results will help to somewhat mitigate the concerns of Wood and Rowland-Morin (1989), Jones (1994), Hunt (1997), and Williams, McGee, and Worth (2001) who warn of the pedagogic and personal tradeoffs when competition becomes the prime motivator for continued participation. At least for this study population, the *debate* subjects did no worse – and, in some cases – significantly better than their *non-debate* peers in the arena of academic success.

Critical Outcome 4 – *Moral/Ethical Issues*

In *Study One*, no significant differences were reported between the subject groups in the areas of *propensity to distort the truth* or a *belief in situational ethics*. *Study One* did report that *debate* students were much less likely to ignore evidence that conflicted with their beliefs and to exhibit a stronger belief in a just society tradition. *Study One* concluded that there appeared to be a stronger belief in the ability of concerned citizens to make a difference on the part of the *debate* study group as compared to their *non-debate* peers. *Study One* and *Study Two* comparisons in each of the four sub-assessments are offered side-by-side and reported in Table 6.

Table 6

8 year Consolidated Comparison between Debate and Non-Debate Groups
Moral / Ethical Outcomes

	<i>Study One</i>		<i>Study Two</i>	
	Undergrad 1997-2001		Grad & Beyond 2001-2005	
	Debate	Non-debate	Debate	Non-Debate
Critical Outcome 4: Moral/Ethical Outcomes				
Propensity to Distort the Truth	No significant Difference		+0.100	+0.236
Belief in Situational Ethics	No significant difference		+0.317	+0.439
Propensity to Ignore Conflicting Evidence	+0.113	+0.817	+0.217	+0.781
Belief in the Just Society Tradition	+0.870	+0.596	+0.787	+0.459

All four comparisons were statistically significant. Whereas *Study One* reported no significant differences between *debate* and *non-debate* subject pools in the assessments *propensity to distort the truth* or *belief in situational ethics*, *Study Two* reflects dramatic changes for both groups. *Non-debate* respondents were more likely to distort the truth and to believe in situational ethics. Modest changes occurred for the *debate* group reflected through a slight rise in the propensity to ignore conflicting evidence and a slight decrease in the belief in a just society tradition.

In summary, *Study Two* demonstrates that during the four-year, post-graduation period the *debate* respondents continued to report statistically significant differences between their behaviors and belief systems and their *non-debate* peers. During the post-graduation period, the positive outcomes reported for undergraduate *debate* participants in *Study One* held somewhat steady or increased. Undergraduate *debate* participants continued to be far less likely to *ignore evidence that conflicted with their beliefs* than their *non-debate* peers as they moved through the post-graduation, graduate programs and professional world. The *debate* group continued to be much more likely to believe in a *just society tradition*. Finally,

though *Study One* reported no statistically significant differences between the two groups, in this follow-up study, the *debate* group was much less likely to *distort the truth* or to *believe in situational ethics*.

Critical Outcome 5 – *Psychological Multipliers*

Study One reported statistically positive behavioral outcomes for those who participated in *debate* in all five areas as compared to their *non-debate* peers. *Debate* participants reported significantly less feelings of depression, anxiety or feelings of being overwhelmed under pressure and significantly higher feelings of confidence and a positive outlook on life. *Debate* participants also reported significantly higher levels of confidence in their ability to communicate their feelings and positions through oral and written communication and were significantly more flexible in their outlook on life. The only area where the study groups exhibited no significant differences was in their ability to maintain successful long-term relationships. Again, *Study Two* sought to confirm the continued relationship between these positive outcomes and debate participation beyond graduation. *Study One* and *Study Two* comparisons are offered side-by-side and reported in Table 7.

Table 7

8 year Consolidated Comparison between Debate and Non-Debate Groups
Psychological Multipliers

	Study One Undergrad 1997-2001		Study Two Grad & Beyond 2001-2005	
	Debate	Non-debate	Debate	Non-Debate
Critical Outcome 5: Psychological Multipliers				
Propensity for Depression/Anxiety	+0.097	+0.413	+0.141	+0.488
Feeling Overwhelmed under Pressure	+0.011	+0.211	+0.107	+0.310
Feelings of Self-Confidence/Positive Outlook	+0.799	+0.700	+0.691	+0.544
Confidence in Communication Skills	+0.873	+0.417	+0.889	+0.579
Maintain long-term relationships	No significant difference		+0.496	+0.276
Propensity for Flexibility	+0.537	+0.336	+0.504	+0.417

During the four-year follow-up period of *Study Two*, *debate* respondents continued to demonstrate stronger relationships than their *non-debate* peers to positive mental health outcomes. *Debate* participants were significantly less likely to report feelings of *depression or anxiety* or feelings of being *overwhelmed under pressure* than their *non-debate* peers. They reported even stronger *feelings of self-confidence* and a *positive outlook* post-graduation, widening the gap between themselves and the *non-debate* group reporting lower levels of confidence and a more negative outlook. The correlated relationship between groups in the assessments *confidence in communication skills* and *flexibility* remained fairly constant, with the *non-debate* group reporting a slight decrease in confidence and a slight increase in *flexibility*. Both measures remained statistically significant in terms of differences in positive outcomes. The assessment measuring the ability of the groups to *maintain long-term relationships* changed dramatically during *Study Two*. At the close of

Study Two the *debate* group was much more successful at *maintaining long-term relationships* than their *non-debate* peers.

In summary of the examination of critical outcome 5) *Psychological Multipliers*, in each of the six critical assessments, the *debate* group demonstrated much higher positive outcomes. In the four years after graduation, the *debate* group was much more likely than their *non-debate* peers to continue to benefit from a much healthier mental outlook through higher overall confidence in themselves and their communication skills, lower levels of depression, anxiety and feelings of being overwhelmed under pressure, a more positive mental outlook, and an ability to more successfully maintain long-term relationships.

One final area of analysis is advanced. In addition to the 84-item survey instrument measuring the five critical outcomes, the subjects were asked to provide responses to a number of questions designed to solicit information pertaining to their professional career experiences. No statistical interpretation of the responses is advanced; thus, some degree of caution regarding any conclusions made from the data is suggested. However, some differences are interesting to note. They are found in Table 8.

Table 8
Career and Professional Choices

	Debate	Non-Debate
Did you have a job offer in your field after graduation?	75% (51 of 68)	55% (28 of 51)
How many times since graduation have you		
changed employment voluntarily?	2.2	1.7
changed employment involuntarily?	.5	1.3
been promoted?	2.3	1.8
experienced an increase in pay?	2.6	2.1
Have you been evaluated by a superior in your job?	92% Yes	94% Yes
Was the evaluation positive?	73% Yes	61% Yes
Was the evaluation negative?	25% Yes	40% Yes
According to the evaluations, what factor(s) contributed most to		
a negative evaluation?	Social Deadlines	Social Poor work product Communication skills
a positive evaluation?	Comm. skills Analytical skills	Work Product Social
Overall, how would you rate your happiness with your career choice?	7.6	6.1

Notes: Social, in the negative sense, meant that the respondent had some difficulty getting along with a co-worker or superior. Social in the non-debate positive sense meant that the respondent was praised for being a team-player or for getting along well with co-workers. Happiness was rated on a scale from 1 to 10 with 1 being low and 10 being high.

In pursuit of an answer to RQ2: *Are there differences in performance between debate and non-debate groups in their career paths post-graduation(?)* some differences in performance are reported. The *debate* group had considerably more job offers in their field of study upon graduation than their *non-debate* peers. They report more often being able to change jobs voluntarily, are forced to change jobs much less, and have experienced a slight edge in terms of receiving an increase in pay. They also experienced proportionately more positive evaluations from their supervisors. When asked to identify key factor(s) that contributed to negative evaluations, both groups reported some difficulty in getting along with a co-worker or supervisor. In addition the *debate* group had some difficulty meeting work product deadlines. The *non-debate* group reported inferior work product and a lack of communication skills. Conversely, when asked to identify key factor(s) that contributed to positive evaluations, the *debate* group cited good communication skills and an ability to think and analyze problems. The *non-debate* group reported that when evaluations were positive, they were most often cited for good work product and their ability to be a team player and/or to get along with co-workers. Therefore, there are at least some differences in performance between groups in their career paths post graduation. Interestingly enough, the *debate* group reports being happier in their various career choices than the *non-debate* group.

Conclusion

Of course, some care should be exercised when making generalized claims from this study sample to the larger *debate* and *non-debate* populations across the United States. The participants who remained in the study for the entire eight-year period displayed above-average commitment levels to the research and, therefore, may not be indicative of larger, less motivated subject pools. Further research with larger study populations is absolutely critical to the proper evaluation of the claims explored through this study. It would also be interesting to follow the respondent group as they continue through life. Initial choices in

career paths and overall performance levels are almost always modified with the passage of time. Additional, more sophisticated research should be undertaken to look for statistical differences in the performances of the groups as they matriculate through life. Again, does undergraduate participation in forensics lead to life-long positive outcomes? Finally, threats to casual conclusions should be explicitly considered before making more generalized conclusions. This difficulty in interpreting the direction of the casual relationships is somewhat mitigated by the longitudinal nature of the study. However, generalized conclusions regarding the casual relationship between debate participation and positive outcomes should be advanced with caution.

In conclusion, the intent of *Study Two* was to provide valid, empirical data to answer the research question: Will significant positive differences in the five critical areas of: academic success; social responsibility; psychological multipliers; cultural tolerance and understanding; and moral/ethical issues as reported in *Study One* continue for the *debate* study population in their post-graduation experiences? The data suggests that, as was concluded in *Study One*, participation in forensics during the study population's undergraduate experience is strongly correlated with increased positive outcomes, in this specific case, beyond graduation. During the additional four years of study, the *debate* group maintained every positive academic, social and behavioral edge reported during the initial study period. In addition, the *debate* group increased the positive outcome gap between themselves and their *non-debate* peers in the assessment areas of: increased propensity to vote; maintaining a GPA above 3.5; being less likely to ignore evidence that conflicts with what they already believe; having greater self-confidence; and demonstrating an increased propensity to engage in social volunteerism and political and social advocacy. Finally, whereas *Study One* reported no statistical differences between groups in several key areas, *Study Two* found that four years later, there were statistically significant differences in the *debate* group's post-graduation experiences, specifically in increased propensity to engage in cross-cultural relationships and to hold membership in cross-cultural organizations, to matriculate on time through graduate and professional programs, and to

maintain long-term relationships. *Debate* respondents were also less likely to distort the truth or to believe in situational ethics. Therefore, *Study Two* concludes that the answer to the research question is that again, in almost every case, in almost every area, forensic participation during the subject population's undergraduate experiences has led to sustained, significant positive life outcomes beyond graduation.

Additionally, there does seem to be at least some evidence that participation in forensics during a student's undergraduate experience does lead to some differences in performance on the job; and therefore, positive options in terms of positive evaluations by superiors, slight increases in the rate of pay raises and promotions, and the ability to move voluntarily from one job to another. Also, those with *debate* experience tend to be involuntarily separated from a job less than their *non-debate* peers. While these findings need further research and support, it would seem safe to conclude that at least for this study's subjects, undergraduate debate participation led to increased professional benefits during the four-year period following graduation.

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